

7 Simplify.

(a) $\frac{p}{t} \div \frac{2}{t}$

$$= \frac{p}{\cancel{t}} \times \frac{\cancel{t}}{2}$$

$$= \frac{p}{2}$$

$$\frac{p}{2} \dots\dots\dots [2]$$

(b) $\frac{3x}{4} - \frac{x-1}{2}$

$$= \frac{3x}{4} - \frac{2x+2}{4}$$

$$= \frac{x+2}{4}$$

$$\frac{x+2}{4} \dots\dots\dots [2]$$

- 8 The cost of one orange is t cents.
The cost of one apple is w cents.

The total cost of 3 oranges and 1 apple is 51 cents.
The total cost of 6 oranges and 5 apples is 129 cents.

Use simultaneous equations to find the value of t and the value of w .
You must show all your working.

orange = t cents

apple = w cents

$$3t + w = 51 - [1]$$

$$6t + 5w = 129 - [2]$$

$$[1] \times 2$$

$$3t + w = 51$$

$$\frac{3t}{3} = \frac{51-9}{3} \quad \therefore \underline{t = 14}$$

$$6t + 2w = 102 - [3]$$

$$[2] - [3]$$

$$6t + 5w = 129$$

$$- \underline{6t + 2w = 102}$$

$$\frac{3w}{3} = \frac{27}{3}$$

$9 = w$

$$t = \underline{14}$$

$$w = \underline{9} \dots\dots\dots [5]$$